SEN307 — LAB 2: DATABASE DESIGN & NORMALIZATION

Paul Kelechi | VUG/SEN/24/12934 | Schema: sen307_VUG_SEN_24_12934

1. UNNORMALIZED FORM (UNF)

The source registration sheet stores multiple course codes, titles, lecturers and lecturer departments in single cells. Example: matric_no 20231234; Ada Obi; Software Engineering; courses SEN307/SEN301; titles Database Systems/Software Architecture; lecturers Dr A. Okafor/Dr M. Bello; session 2026/2027. These repeating groups cause non-atomic values and ambiguity in row-level associations.

2. FIRST NORMAL FORM (1NF)

Split repeating course values into one student–course row per course. Keep each cell atomic. Candidate composite key: (matric_no, course_code).

matric_no	student_name	student_dept	course_code	course_title	session
20231234	Ada Obi	Software Engineering	SEN307	Database Systems	2026/2027
20231234	Ada Obi	Software Engineering	SEN301	Software Architecture	2026/2027
20231235	John Musa	Computer Science	SEN307	Database Systems	2026/2027
20231235	John Musa	Computer Science	COS301	Operating Systems	2026/2027
20231236	Grace Okon	Software Engineering	SEN301	Software Architecture	2026/2025

Fix: repeating groups are now separate rows, but student and course facts are duplicated across the composite key. The lecturer lists are not repeated in these rows because the source does not unambiguously associate each lecturer with each listed course.

3. SECOND NORMAL FORM (2NF)

Student(matric_no PK, student_name, student_dept); **Course**(course_code PK, course_title); **Enrolment**(matric_no PK/FK, course_code PK/FK, session). Student details depend only on matric_no; course title depends only on course_code; the enrolment key represents a student-course registration.

4. THIRD NORMAL FORM (3NF)

Remove the transitive department dependency by creating Department and linking both Student and Lecturer to it. Represent lecturer teaching assignments by CourseOffering, which identifies a course, lecturer and session. Enrolment links a student to a specific offering. This supports the stated rule that courses can have multiple lecturers across sessions.

Table	Columns and keys
Department	dept_id **PK**, dept_name UNIQUE
Student	student_id **PK**, matric_no UNIQUE, full_name, dept_id **FK** → Department
Lecturer	lecturer_id **PK**, lecturer_name, dept_id **FK** → Department
Course	course_id **PK**, course_code UNIQUE, course_title
CourseOffering	offering_id **PK**, course_id **FK** → Course, lecturer_id **FK** → Lecturer, session; UNIQUE(course_id, lecturer_id, session)
Enrolment	(student_id, offering_id) composite **PK**; both columns are **FKs** to Student and CourseOffering

5. RELATIONSHIPS & CARDINALITIES

Department 1:M Student; Department 1:M Lecturer; Course 1:M CourseOffering; Lecturer 1:M CourseOffering; Student 1:M Enrolment; CourseOffering 1:M Enrolment. Therefore Student and CourseOffering are M:N through Enrolment. A course can have multiple lecturers across sessions through CourseOffering.

6. DESIGN NOTE

The supplied sheet lists lecturers and lecturer departments as multi-valued cells but does not reliably pair each lecturer with each course/session. The design therefore provides CourseOffering to store confirmed assignments without inventing sample mappings. The schema and ERD are aligned; see lab2_schema.sql and lab2_erd.png.